



SECI1143

PROBABILITY & STATISTICAL DATA ANALYSIS

SECTION 2

PROJECT 2

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Table of Contents

1.0 Introduction	3
2.0 Dataset	3
2.1 Description of Data	3
2.2 Statistical Test Analysis	4
3.0 Data Analysis	5
3.1 Hypothesis Test with Two Sample (Two Mean)	5
3.2 Correlation Test	6
3.4 Chi-square test of Independence	9
4.0 Conclusion	10
5.0 Appendix	11

1.0 Introduction

Heart attack, or myocardial infarction, is a critical health condition that occurs when the blood flow to a part of the heart is blocked for an extended period, leading to the damage or death of heart muscle tissue. Statistically, approximately 18 million people globally succumb to cardiovascular diseases annually, with heart attacks accounting for a significant portion. But what are the actual factors that contribute to the diagnosis of heart attack disease?

In project 2, we picked datasets from websites such as Kaggle, Data.gov, Google's dataset, and DataHub. This project has a few compulsory criteria, including hypothesis testing using one or two samples, as well as correlation and regression. We selected the most appropriate dataset from Kaggle, "Heart Attack Risk Prediction Dataset". This secondary data was obtained by Dr. Sourav Banerjee, a professor in the Department of Mechanical Engineering at the University of South Carolina.

The goal of this project is to apply and perform statistical test analysis on secondary data source and to see if there is a link between the selected variables from the data set. Our group has opted to utilize several approaches to analyse and interpret the data set, including hypothesis testing with two samples, correlation testing, regression testing, and the chi-square test of independence. We hope that by doing these experiments, we will be able to confirm or contradict our expectations.

2.0 Dataset

2.1 Description of Data

Population: Heart attack disease patient around the globe

Sample: 8763 patients

This dataset is collected from online sources (Kaggle). This secondary data consists of 26 variables, but we only choose 7 out of 26, which are age, gender, heart rate, BMI, cholesterol, smoker and heart attack risk. The table below shows the details of each variable used regarding their level of measurement and type of variable.

No	Variable / Parameter	Level of Measurement	Type of variable
1.	Age	Ratio	Quantitative, Discrete
2.	Gender	Nominal	Qualitative, Categorical
3.	Heart rate	Ratio	Quantitative, Discrete
4.	BMI	Ratio	Quantitative, Continuous
5.	Cholesterol	Ratio	Quantitative, Discrete
6.	Smoker	Nominal	Qualitative, Categorical
7.	Heart attack risk	Nominal	Qualitative, Categorical

2.2 Statistical Test Analysis

Selected Variables	Objectives	Test Analysis and Expected Outcome
Gender, heart rate	To investigate if there is a difference between heart rate of male and female patients.	Analysis: Hypothesis 2-sample testing (Test on Mean, Variance Unknown) Expected Outcome: By conducting a 2-sample t-test, we can compare the mean heart rate between male and female patients. If the $p\text{-value} < 0.05$, it would suggest that there is a significant difference in heart rate between the two groups.
Age, heart rate	To investigate the relationship between age and heart rate.	Analysis: Correlation analysis Expected Outcome: By calculating the correlation coefficient between age and heart rate, we can determine the strength of the relationship. If the correlation coefficient is positive and statistically significant, it means as age increases, heart rate tend to be higher.
Cholesterol, BMI	To provide insight if cholesterol is a predictor of BMI.	Analysis: Simple linear regression Expected Outcome: By using a simple linear regression test, relationship between cholesterol and BMI can be determine. If the regression coefficient < 0.05, it would suggest we can quantify the extent to these variables predict the likelihood of experiencing a heart attack.
Smoker, heart attack risk	To investigate the independence between smoker and risk of heart attack.	Analysis: Chi-square test of Independence Expected Outcome: By using a chi-square test of independence, we can know the relationship of smoker and heart attack risk. If the $p\text{-value}$ of chi-square test < 0.05, it would suggest that the smoker and heart attack risk are dependent.

3.0 Data Analysis

3.1 Hypothesis Test with Two Sample (Two Mean)

We have use variables Gender and Heart Rate in this section to determine is there any difference between mean Heart Rate of female and mean Heart Rate of male. We conduct the t-test with a significant level, $\alpha = 0.05$ and assume unequal variances. By using R, data has been calculated.

	Female	Male
Frequency, n	n1 = 2652	n2 = 6111
Mean, $\bar{x}$	$\bar{x}_1 = 75.4$	$\bar{x}_2 = 74.9$
Standard Deviation, s	s1 = 20.6	s2 = 20.5

Hypothesis statement

$H_0: \mu_1 = \mu_2$

$H_1: \mu_1 \neq \mu_2$

μ_1 represent mean for female patients while μ_2 represent mean for male patients.

Test statistic

We have 2 options to calculate test statistics, T_0 including formula and using R.

By using formula : $t_0 = \frac{\bar{x}_1 - \bar{x}_2 - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$

By using RStudio, the value of t_0 is 1.0197. The p-value is 0.3079. Then, we can calculate degree of freedom, v with the formula or using R.

By using formula : $degree\ of\ freedom, v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$

By using RStudio, the value of v is 5010.376. We assume $v \approx 5010$.

For critical value, we can only calculate through RStudio as the value of v is too large, it is not included in t - table. $\alpha = 0.05$. Since this is a two-tailed test, there are two critical values which are lower tail critical value, $t_{0.025, 5071} = -1.96$ and upper tail critical value, $t_{0.025, 5071} = 1.96$.

Discussion

Since $t_0 = 1.04 < t_{0.025, 5071} = 1.96$, **failed to reject** null hypothesis. There is insufficient evidence to support the claim that mean heart rate of female is different with mean heart rate of male at a significant level of 0.05. So, gender is not a main indicator of the heart rate. Factors such as high variability in heart rate measurements, similarities in biological and physiological characteristics and unaccounted lifestyle might have contributed to this finding.

3.2 Correlation Test

In this analysis, we aim to examine the relationship between age as independent variable (x) and heart rate as dependent variable (y) in a sample of 8,763 (n) individuals using Pearson's Product-Moment Correlation Coefficient to calculate the correlation coefficient, r at a 95% confidence level. The formulas for calculating the sample correlation coefficient and test statistic formulas are shown below.

$$\text{sample correlation coefficient, } r = \frac{\Sigma xy - (\Sigma x \Sigma y)/n}{\sqrt{[(\Sigma x^2) - (\Sigma x^2)/n][(\Sigma y^2) - (\Sigma y)^2/n]}}$$

$$\text{Test statistic formula, } t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}; \text{ degree of freedom, } df = n - 2$$

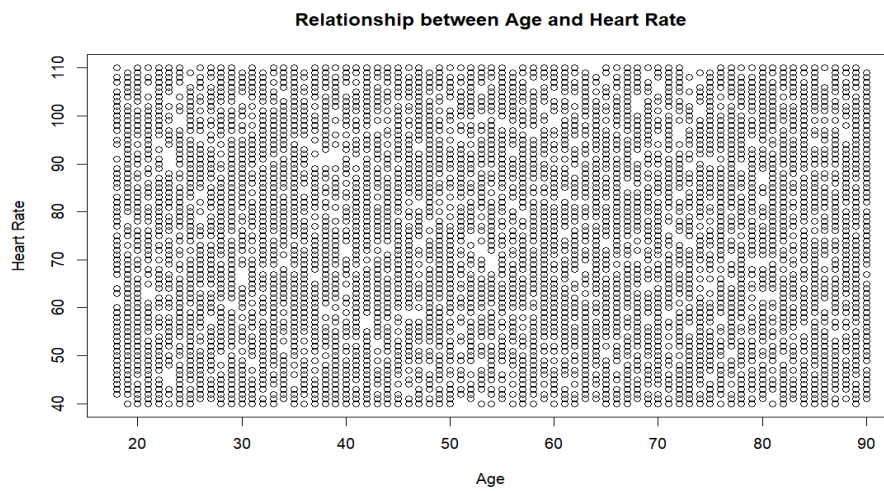


Figure 3.1 Visualise data in scatter plot

From the scatter plot in Figure 3.1, we can see that the data points are evenly spread without any clear pattern and trend, indicates that there is an extremely weak negative linear relationship between age and heart rate. In the other words, the changes in age do not correspond to consistent changes in heart rate levels.

Hypothesis Statement

$H_0: \rho = 0$ (no linear correlation)

$H_1: \rho \neq 0$ (linear correlation exists)

Where ρ represent the population correlation coefficient.

Test statistic

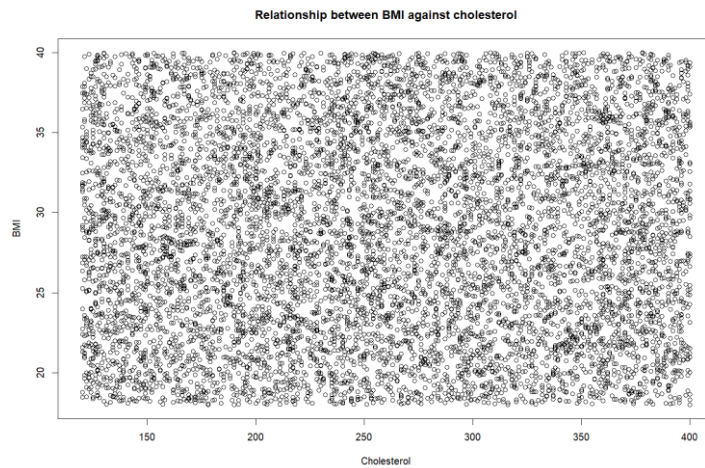
By using Rstudio, the sample correlation coefficient, r is calculated as -0.003844013 , test statistic value, t is -0.3598031 and p-value is 0.719 . At 95% confidence level ($\alpha = 0.05$) and degree freedom of 8761 is used to find the critical value. Since this is a two-tailed test, there are two critical values which are lower tail critical value $t_{0.025, 8761} = -1.96$ and upper tail critical value $t_{0.025, 8761} = 1.96$.

Discussion

Since $t = -0.3598031$ are in the range between lower tail critical value, $t_{0.025, 8761} = -1.96$ and upper tail critical value, $t_{0.025, 8761} = 1.96$, $t_{0.025, 8761} = -1.96 < t = -0.3598031 < t_{0.025, 8761} = 1.96$, it is **failed to reject** the null hypothesis. There is not enough evidence to support the claim that linear correlation exists between the age and heart rate of the patients, at a significant level of 0.05. These data may not support a significant linear relationship between age and heart rate due to factors such as individual variability, lifestyle differences, health conditions, medication use and others more. These factors indicate that other influences besides age may play a more significant role in determining heart rate.

3.3 Regression Test

In this analysis, we aim to investigate whether BMI is influenced by cholesterol levels. Here, cholesterol serves as the independent variable (x), while BMI is the dependent variable (y). We will employ a simple linear regression model to assess this relationship, operating under the assumption that variations in cholesterol levels drive changes in BMI.



The scatter plot shows a wide dispersion of data points with no clear trend, indicating a weak or non-existent linear relationship between cholesterol levels and BMI. The lack of a discernible pattern suggests that variations in cholesterol levels do not consistently influence BMI.

Estimated Regression Model

$$\hat{y}_i = b_0 + b_1x$$

In the equation, b_0 is the estimated average BMI when cholesterol is zero, known as the y-intercept. b_1 represents the estimated change in average BMI for each one-unit increase in cholesterol. By using RStudio, we obtained the value of $b_0 = 253.484$ and $b_1 = 0.221$. As we know b_0 in the equation represents y-intercept, hence estimated intercept is equal to b_0 . BMI represents the value of b_1 . Substitute the values of b_0 and b_1 into the regression model equation:

$$\hat{y}_i = 253.484 + 0.221x$$

The equation provides insights into the relationship between cholesterol and BMI. The intercept value of 253.484 represents the estimated average BMI when cholesterol is zero, though this is biologically unrealistic. The slope value of 0.221 indicates that for each one-unit increase in cholesterol, BMI is expected to increase by 0.221 units on average. This positive slope suggests a correlation between higher cholesterol levels and higher BMI values. While the intercept lacks practical significance, the slope meaningfully illustrates how BMI tends to change with cholesterol levels.

❖ Explained and Unexplained Variation:

By using RStudio, we get $SSE = 57276442.591$, $SSR = 17131.288$ and $SST = 57293573.879$.

❖ Find Coefficient of Determination, R^2

By using RStudio, we get coefficient of determination, $R^2 = 0.0003$. Hence, R^2 value of 0.0003 indicates that a very weaker linear relationship between cholesterol and BMI. This is because there is only a very small fraction (0.03%) of the variation in BMI can be explained by variations in cholesterol levels according to the linear regression model.

❖ Find Standard Deviation of Regression Slope using the formula:

By using RStudio, we get the estimate of the standard error of the least squares slope, $S_{b1} = 0.1367$. The value of S_{b1} suggests that there is an estimated standard deviation of approximately 0.1367 units around the estimated slope coefficient b_1 . This means that the actual slope coefficient in the population is likely to be within approximately ± 0.1367 units of the estimated value from our sample data.

Hypothesis Statement

$H_0: \beta_1 = 0$ (Cholesterol has no linear relationship with BMI)

$H_1: \beta_1 \neq 0$ (Cholesterol has linear relationship with BMI)

β_1 represent the population slope coefficient of cholesterol with BMI.

Test Statistic

We will find the critical value by using $\alpha = 0.05$, $df = n - 2 = 8761$. Since this is a two-tailed test, two critical values gotten from RStudio are $-t_{0.025, 8761} = -1.96$ and $t_{0.025, 8761} = 1.96$. Hence, we reject H_0 if test statistics < -1.96 or test statistics > 1.96 . By using RStudio, we get the test statistic $t_0 = 1.619$. The p-value for $t_0 = 1.619$ are 0.1055.

Discussion

Since the test statistic $t_0 = 1.619$ is less than $t_{0.025, 8761} = 1.96$, we **failed to reject** the null hypothesis. There is insufficient evidence to support the claim that cholesterol has linear relationship with BMI at a significant level of 0.05. BMI is influenced by various factors such as diet, physical activity, genetics, and health conditions. Cholesterol, though related to diet and lifestyle, may not significantly impact BMI as BMI measures body fat based on height and weight, without accounting for factors like muscle mass.

3.4 Chi-square test of Independence

In this study, we will analyse the relationship between **smoking** status (smoker) and the **risk of heart attack** using a Two-Way Contingency Table at a 95% confidence level. To determine if there is a statistically significant association between smoking status and the risk of heart attack, we will use the Chi-Square Test of Independence alongside the contingency table analysis.

Hypothesis Statement

H₀: Both variables are independent

H₁: Both variables are dependent

Test Statistic

By using RStudio, we get critical value, $X^2 = 5.99$ and $df = 2$.

1. Calculate the expected counts:

Smoking	Risk of heart attack				Observed total
	0		1		
	Observed	Expected	Observed	Expected	
0 (No)	575	$\frac{904 \times 5624}{8763}$ = 580.178	329	$\frac{904 \times 3139}{8763}$ = 323.822	904
1 (Yes)	5049	$\frac{7859 \times 5624}{8763}$ = 5043.822	2810	$\frac{7859 \times 3139}{8763}$ = 2815.178	7859
Total	5624	5624.103	3139	3139	8763

2. Calculate the test statistic value:

➤ Calculated manually

Cell, ij	Observed count, o _{ij}	Expected count, e _{ij}	$\frac{o - e}{e}$
1, 1	575	$\frac{904 \times 5624}{8763} = 580.178$	$\frac{(575 - 580.178)^2}{580.178} = 0.046$
1, 2	5049	$\frac{7859 \times 5624}{8763} = 5043.822$	$\frac{(5049 - 5043.822)^2}{5043.822} = 0.005$
2, 1	329	$\frac{904 \times 3139}{8763} = 323.822$	$\frac{(329 - 323.822)^2}{323.822} = 0.083$
2, 2	2810	$\frac{7859 \times 3139}{8763} = 2815.178$	$\frac{(2810 - 2815.178)^2}{2815.178} = 0.010$
χ^2			0.144

When we calculate test statistic manually, we get test statistic, $X^2 = 0.144$

➤ Using RStudio

When we calculated test statistic using RStudio, we get the same value as calculated manually which is $X^2 = 0.14383$ with p-value of 0.7045.

Discussion

Since the test statistic value $X^2 < X^2_{k=1, \alpha=0.05}$, $0.144 < 5.991$, we **failed to reject** the null hypothesis. Thus, there is insufficient evidence to support the claim that both variables, smoking status and heart attack risk are dependent at a 95% confidence level. Variations in individual responses to smoking, smoking habits, and other lifestyle factors may contribute to the non-significant result, highlighting the need for comprehensive risk assessments for heart attack likelihood.

4.0 Conclusion

In a two-sample hypothesis test comparing the mean heart rates of female and male patients, we found that the mean heart rates for both groups were the same. Consequently, we **failed to reject** the null hypothesis. Therefore, we conclude that gender is not a significant indicator of heart rate. Factors such as high variability in heart rate measurements, similarities in biological and physiological characteristics, and unaccounted lifestyle differences might have contributed to this result.

For the correlation test, our research suggested that as age increases, heart rate tends to be higher. However, we **failed to reject** the null hypothesis again. The data may not support a significant linear relationship between age and heart rate. Factors such as individual variability, lifestyle differences, health conditions, medication use, and other influences may play a more significant role in determining heart rate than age alone.

Next, we considered the possibility that BMI might be influenced by cholesterol levels for regression test. Based on our linear regression model equation, $\hat{y}_i = 253.484 + 0.221x$, we **failed to reject** the null hypothesis. This outcome is likely due to individual differences in metabolism and how the body processes and stores fat, which can vary greatly, leading to high variability in BMI that cannot be solely explained by cholesterol levels. The complexity and the presence of multiple contributing factors may dilute the direct correlation between cholesterol and BMI, resulting in a non-significant statistical relationship in the analysis.

Lastly, we analyzed the relationship between smoking status (smoker) and the risk of heart attack using the chi-square test of independence. We **failed to reject** the null hypothesis. The lack of significant evidence to support a relationship between smoking status and the risk of heart attack can be attributed to several factors. While smoking is a known risk factor for heart disease, heart attacks can also be influenced by genetic predisposition, diet, physical activity,

stress, underlying medical conditions, and medication use. The presence of these various factors may overshadow the impact of smoking alone in this particular study sample.

In conclusion, the likelihood of a heart attack is not significantly influenced by the patient's gender, age, or smoking status. Additionally, BMI does not appear to have a significant impact on cholesterol levels. These findings suggest that other factors, beyond the ones studied, may play a more crucial role in determining heart attack risk and cholesterol levels.

We have successfully conducted various analyses including two-sample hypothesis testing, correlation analysis, regression analysis, and the chi-square test of independence using RStudio. This project has been invaluable in enhancing our data analysis skills for future applications. We would like to extend our gratitude to Dr. Noorfa, our lecturer, for her invaluable support and guidance throughout this endeavour.

5.0 Appendix

Video Presentation: https://youtu.be/o_1EMPVodgg

Portfolio Overall Reflection & Video : https://www.linkedin.com/posts/pohlkeyee_reflecting-on-probability-and-statistical-activity-7214192091529191425-CAp-?utm_source=share&utm_medium=member_desktop

Raw Data Set : <https://www.kaggle.com/datasets/iamsouravbanerjee/heart-attack-prediction-dataset>